AWI–ESM3 v3.4 PI-control tuning campaign — summary

Where the three objectives stand, and how to make the Southern Ocean and Siberia fit together

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50 atmosphere-only lever runs, 28 coupled runs

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Abstract

This is the short reading of a campaign whose full record runs to eighty-six pages (`report.tex`). It keeps the results that still constrain a decision and drops the mechanisms that died on the way. Three objectives are tracked — the **global radiative imbalance**, the **Southern Ocean**, and the **Siberian cold bias** — and the campaign's central operational finding is about the relationship between them.

The headline. Across 29 coupled runs the two original faults trade off almost perfectly linearly: every $\mathrm{W\,m^{-2}}$ of TOA improvement has cost ≈0.6 K of global cold. That line is *not* a property of the model. It is a property of the one lever family that was used to buy the imbalance — aggressive ocean mixing, which lowers TOA by cooling the planet. The regime-selective levers built later escape it: D2b (Southern-Ocean ice nucleation) and F4/G4 (boreal surface exchange) act on disjoint geography and disjoint physics, and when combined they superpose additively to within noise on every metric. **Southern Ocean and Siberia are not in competition; the ocean-mixing stack was the competition.**

Where we are. K1 + P5 is carried forward: Siberian JJA +1.15 K with every season inside its own noise threshold, on a snow scheme rebuilt from 36 492 Russian snow-course surveys after the original was falsified outright. Coupled 10A sits at net TOA +0.85 $\mathrm{W\,m^{-2}}$ against a goal of 0, of which only 0.18 is still reachable from the ocean side. The Southern Ocean carries +5.81 $\mathrm{W\,m^{-2}}$ of band error from a 6.6 pp cloud-*area* deficit that no lever has yet moved by more than a fraction.

What to do next is §8: attack the SO cloud-area deficit rather than escalating ocean mixing; stop optimising the snow scheme; and diagnose the global tropospheric cold bias, which is now the shared limiter on both remaining objectives and has never been worked on in 50 runs.

1 Scorecard

Coupled numbers are 26-yr means on the common window 1354–1379; AMIP numbers are 44-yr means over 1872–1915 scored by run×year ANOVA. **ERA5 and CERES are present-day and the production runs are pre-industrial**; where that matters the period-clean value is given.

objective	metric	start	now	target	what moved it
Global radiation	coupled net TOA [$\mathrm{W\,m^{-2}}$]	1.73	**0.85**	0	ocean mixing, sea ice, G4
	of which atmospheric	0.67	0.67	0	*nothing*
	of which coupled-state	1.06	**0.18**	0	nearly exhausted
	period-clean vs CERES	+1.23	+1.23	0	*never targeted*
Southern Ocean	65–45S net TOA vs CERES	+5.81	+5.81	0	—
	SO surface-SW RMSE	6.88	**4.81**	—	D2b (INP)
	SO cloud area [%]	83.1	83.6	89.7	6.1 pp untouched
Siberia	JJA T2m bias, period-clean [K]	−2.04	≈**−0.9**	0	G4 + K1 + P5
	JJA T2m gain [K]	—	**+1.15***	—	+1.04 K1, +0.11 P5
	DJF penalty [K]	—	**−0.06**	$\lvert\cdot\rvert < 0.588$	clean
	global T2m RMSE (AMIP) [K]	1.579	**1.524**	—	K1

Two entries deserve emphasis because they reframe the whole campaign. The **atmospheric floor of +0.67 $\mathbf{W\,m^{-2}}$ has not moved at all** — it is what the atmosphere does over an observationally correct ocean, so it cannot be ocean heat uptake, and it is now almost the entire coupled imbalance. And the **period-clean radiative error is +1.23, not +0.64**: the present-day arm of the same model reads net TOA +2.20 against CERES's +0.97, so scoring a PI run against a zero target *understates* the radiative fault by about a factor of two.

2 The global radiative imbalance

2.1 The target is zero, and the reason is snow

The inherited goal of $-0.16\,\mathrm{W\,m^{-2}}$ was never derived; it is whatever the HR piControl settled at. A target displaced from zero is defensible only if the atmosphere fails to conserve energy: with a spurious atmospheric source S, equilibrium gives SFC = TOA + S, so zero *ocean* drift would require TOA = $-S$. Measured on 10A over 1390–1399:

term	$\mathrm{W\,m^{-2}}$
`ssr+str+sshf+slhf`	+1.843
snow enthalpy (`sf` $\cdot\, 3.3355 \times 10^8$)	−0.966
net surface	**+0.877**
net TOA	**+0.854**
residual	**−0.023**

The atmosphere conserves to $0.02\,\mathrm{W\,m^{-2}}$, so "the ocean is not drifting" and "the system is not drifting" are the same condition and both put the goal at **0**. **Snow enthalpy does nearly all the reconciliation** — 0.97 of a 0.99 discrepancy. Omit it and the surface reads +1.84 against a TOA of +0.85; that artefact is exactly the kind of apparent leak an offset target gets invented to absorb. Anyone comparing a raw surface flux against TOA without it will see a spurious $\approx 1\,\mathrm{W\,m^{-2}}$ gap.

2.2 Where the imbalance lives: nowhere, and then somewhere

Decomposing the AMIP +0.67 against CERES-EBAF was expected to localise it and appeared not to: global cloud radiative effect matches observation to $0.08\,\mathrm{W\,m^{-2}}$ and planetary albedo to 0.002. That conclusion was an artefact of comparing bands to *each other*. Differenced band-by-band *against CERES*:

band	model	CERES	diff	area	**contribution**
90S–65S	−97.0	−104.0	+7.03	4.8 %	**+0.34**
SO 65S–45S	−49.5	−55.3	**+5.81**	9.8 %	**+0.57**
45S–30S	−5.6	−1.4	−4.20	10.4 %	−0.44
tropics 30S–30N	+42.6	+45.1	−2.51	50.0 %	−1.26
30N–45N	−4.8	−5.7	+0.89	10.4 %	+0.09
45N–65N	−50.1	−54.1	+4.02	9.8 %	+0.40
65N–90N	−106.3	−106.1	−0.15	4.8 %	−0.01

The global budget is right by cancellation, not by being right. Two large regional errors of opposite sign — a Southern Ocean absorbing $+5.8\,\mathrm{W\,m^{-2}}$ too much and tropics −2.5 too little — nearly annihilate. This is why no global knob has ever helped, and why the two largest regional errors having opposite sign is what makes a regionally targeted strategy possible at all.

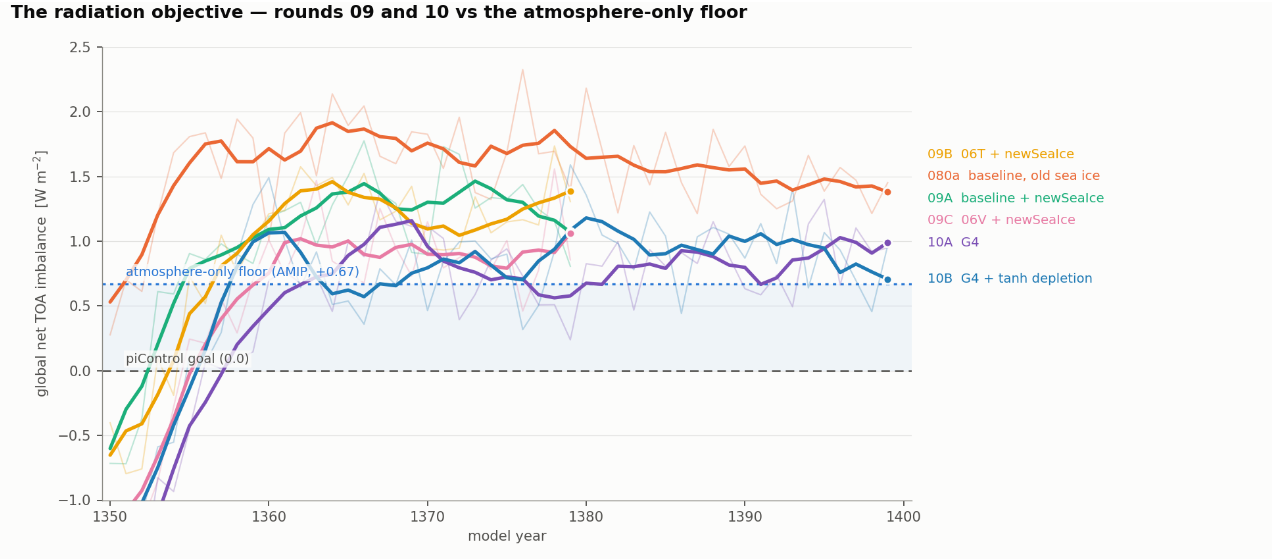

Figure 1: Global-mean net TOA by model year; thin annual, bold 5-yr running mean. **G4 is worth ≈0.25 $\mathbf{W\,m^{-2}}$ at top of atmosphere** — 10A settles at +0.854 against 09C's ≈1.06, the best of the previous round. The dotted line is the **atmosphere-only floor** of +0.67 measured directly by AMIP; the shaded band is the only part of the remaining gap that ocean-side tuning can still close, and everything above the dotted line is atmospheric. **10A and 10B are indistinguishable** (+0.854 vs +0.899) although 10B carries the snow scheme that destroyed the winter soil by 20 K — distributional surface damage is invisible to box-mean energy diagnostics, so TOA cannot be used as a safety check on a surface parameterisation.

2.3 What is left, per run

	net TOA	= atmosphere +	coupled-state
080a (untuned coupled)	1.38	0.67 +	**0.71**
09C (06V + newSeaIce)	1.06	0.67 +	**0.39**
10A (+G4)	**0.85**	0.67 +	**0.18**

The ocean-side lever family is spent. It has delivered three quarters of what was ever available to it, and 0.18 $\mathrm{W\,m^{-2}}$ remains — against a 0.67 atmospheric term that nothing in the campaign has touched. Any further imbalance work must be atmospheric, and §3 says where.

3 The Southern Ocean

3.1 It is a cloud deficit, and it is atmospheric

The SO band's +5.81 $\mathrm{W\,m^{-2}}$ is not subtle in origin: cloud area 83.1 % against CERES's 89.7 % (−6.6 pp), SW CRE −60.3 against −68.1 (+7.8 $\mathrm{W\,m^{-2}}$ too little reflection). Measured in AMIP, *with SST and sea ice prescribed to observations*, so it cannot be an ocean-state error. **This is precisely the Hyder et al. (2018) signature** — the leading 40–60°S warm bias traced to atmospheric net-surface-flux errors dominated by shortwave and cloud.

The measurement is also period-clean: the present-day arm gives an SO SW CRE gap of +7.78 against the PI arm's +7.85, an epoch offset of only −0.07 $\mathrm{W\,m^{-2}}$. Whatever this is, it is the same in both epochs.

3.2 The compensating-error problem, stated plainly

Part I of the campaign bought most of its imbalance reduction from ocean mixing — 1-hourly coupling, MOMIX, weaker KPP — which suppresses spurious Weddell/Ross open-ocean winter convection, thickens the pack, and removes the SO surface warm bias. Damping that convection is physically the right direction: it is the canonical CMIP Antarctic bias, and models over-express a real-but-rare mode. But three independent lines now say the lever is partly compensating for an atmospheric error:

- **Our own AMIP measurement.** The SO carries +5.81 $\mathrm{W\,m^{-2}}$ with the ocean held at observations. That component cannot be fixed from the ocean side, and an ocean lever that removes the resulting surface warmth is cancelling a symptom.

- **The CMPI region map.** 06O/06T improve `thetao`, `so` and `siconc` — exactly where ocean mixing should act — but degrade `tas`/`zg`/`rlut` by +0.16 to +0.26 and redden *every* region including the tropics and Niño-3.4. Cooling leaking to where the parameter has no business acting is the diagnostic signature of a compensating error.
- **The price.** 0.6 K of global cold per W m^{-2} (§5).

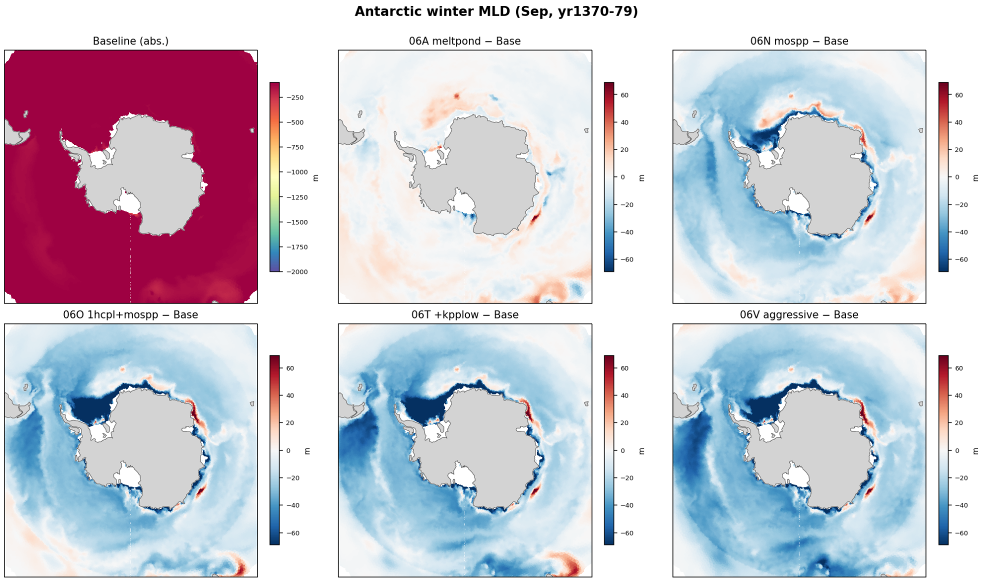

Figure 2: Antarctic September mixed-layer depth: Baseline absolute (top-left, own scale), others anomaly vs Baseline on ±70 m. The mixing runs (06N/O/T/V) restructure Weddell/Ross winter convection; melt-pond (06A) barely touches it. The mechanism is real and the direction is right — the objection in §3 is to how much of the *radiative* target is being bought this way, not to the ocean physics.

3.3 What has actually moved it, and what has been excluded

- **D2b (ice nuclei over sea, gated below 700 hPa) is adopted.** Best SO surface-SW RMSE of any run, 6.88 → 4.96 (−28 %), at *zero* tropical cost (+0.09). It beats the ungated D2a despite a smaller mean-CRE change, because it improves the spatial *pattern*. The pressure gate did exactly what it was designed to do: D2a's tropical cost vanishes while 55 % of the SO gain is retained.
- **Independent support for the mechanism.** SO relative humidity is +5.8 to +6.6 % too high through the mixed-phase layer *while* reflecting too little shortwave — humidity present, liquid absent, which is the supercooled-liquid deficit an INP error predicts.
- **Aerosol is excluded, by measurement.** It had been *inferred* at ∼0.7 W m^{-2} (13 % of the SO clear-sky excess) from a +0.033 AOD excess against MISR. Removing *all* anthropogenic MACv2-SP plumes moves SO clear-sky by **−0.116** W m^{-2} ($t = -18.9$, paired over 25 years) — **six times smaller**, ∼2 % not 13 %. The unexplained residual therefore *grew*, from ∼1.3 to ∼1.9 W m^{-2}. A related trap: M1's effect is overwhelmingly on *cloud* (−0.54 global against +0.13 clear-sky, via `LMACV2SP_CCNF`), so any future aerosol argument scored on the clear-sky column alone is invalid.
- **The 60–90S band is the most responsive in the model**, moving −0.4 to −3.3 under levers designed for entirely different purposes. It is not inert; it has simply never been targeted. The G-series buys about −2.0 of the +8.4 needed there, all of it from D2b, as a by-product.

Net: roughly a quarter of the Southern Ocean gap is closed, all of it incidentally, and the ∼6 pp cloud-area deficit remains untouched by all 34 evaluated levers. That is the single largest unexploited term in the campaign.

4 Siberia

4.1 The bias, and its two routes

The period-clean Siberian JJA cold bias is −2.04 K (measured directly against the model's own present-day arm, not chained through observed epoch change — the earlier indirect estimate was wrong by 2.7×). AMIP alone reproduces essentially all of the coupled model's summer bias (−2.16 against 09A's −2.19), so **the summer half is atmospheric**; the cold-season half is not, and belongs to the coupled vegetation problem.

Two routes were identified and both are now resolved:

Route A — transpiration to cloud: closed. Siberian JJA cloud fraction was 78.1 % against CERES's 69.6 %, carrying $-7.04\,\mathrm{W\,m^{-2}}$ of the surface SW deficit. Raising minimum canopy resistance on boreal types (F4, then G4 adding tundra) moves the cloud term $-7.04 \rightarrow +1.20\,\mathrm{W\,m^{-2}}$, worth +0.952 K, with the tropics untouched (+0.12). Bowen ratio moves 0.44 → 0.60, into the observed range without overshooting. *Caveat kept:* `RVRSMIN` 250 → 1000 is an unanchored 4× excursion from what ECMWF ships, and it does not saturate — 500/1000/2000 give $+0.336/+0.521/+0.876$, so 1000 is a cap set by winter damage, not a knee.

Route B — snow to albedo: real, measured, and worth far less than it looked. June surface albedo is +0.046 too high against CERES, June snow cover 0.380 against a satellite 0.203, melt-out 13 days late. The cause was found: HTESSEL has *no* sub-grid snow depletion, so cover stays pinned near 1 while the median cell still holds 27 cm in May. A depletion scheme was built, and that took five rounds to get right (§4.2). The honest verdict on the whole line is +0.1 to +0.3 K of Siberian JJA, not the ≈1 K the early rounds implied.

4.2 The snow scheme, in one paragraph

The first formulation used tanh. Its range is open at 1, so complete snow cover is *unrepresentable at any parameters* — while Russian snow courses report exactly 10/10 in 99.74 % of 14 369 Siberian DJF surveys. Cost, against 174 676 station soil observations: −20.2 K, where the scheme *off* is right to +0.8 K. The damage was distributional, not a matter of mean cover: DJF cover is identical to 0.002 across the variants while the soil differs by 23 K, because a clipping ramp leaves 93.5 % of the box perfectly insulated and the tanh smears the same deficit into every cell. It was rebuilt as $\mathrm{SCF} = (1 - c_{vh}) \min(1, (d/d_{cl})^{b_l}) + c_{vh} \min(1, (d/d_{ch})^{b_h})$ with $d_c \propto (\rho/\rho_{ref})^{4.7}$ — density, a prognostic the model already carries, as the age/melt proxy, so the scheme works in a 10 °C warmer world or an ice age — fitted to 36 492 snow-course surveys, then floored by a minimum snow *mass* `SWEMIN` after the literal fit gave full cover to a 0.56 mm dusting and aborted the run. **P5 (`SWEMIN`= 15) is adopted**: every season inside its own threshold, and the autumn cover excess the fit introduced cut by 40 % against Rutgers.

run	DJF	MAM	JJA	SON
K1 (base)	−0.496	−0.265	+1.036*	+0.238
N2 `tanh` (falsified)	−2.783*	−0.599*	+1.350*	−0.888*
P3 `SWEMIN`= 3	−0.584	−0.333	+1.341*	+0.217
P5 `SWEMIN`= 15	**−0.060**	**−0.140**	**+1.149***	**+0.004**
P6 `SWEMIN`= 30	−0.850*	−0.288	+1.096*	+0.487*

* clears that season's own 44-yr threshold (DJF ±0.588, MAM ±0.386, JJA ±0.242, SON ±0.431 — a factor 2.4 across the year, and applying the JJA threshold to a winter delta overstates significance by that much).

4.3 The ceiling, and what it exposes

Marginal over K1 against the ±0.242 threshold: P3 +0.305 (marginal), P5 +0.113 (noise), P6 +0.060 (noise). The reason is structural. Against Rutgers' observed May cover of 0.655, the as-released model reads 0.773 — **the entire available error is** 0.118, P5 removes 0.044, and a perfect correction would be worth roughly +0.2 K. The tanh's larger apparent gain was never legitimate: it took 0.158 out of a 0.118 error, overshooting into a −0.040 deficit, so any comparison against its +1.350 is a comparison against an error. **Further `SWEMIN` optimisation is not worth model time.**

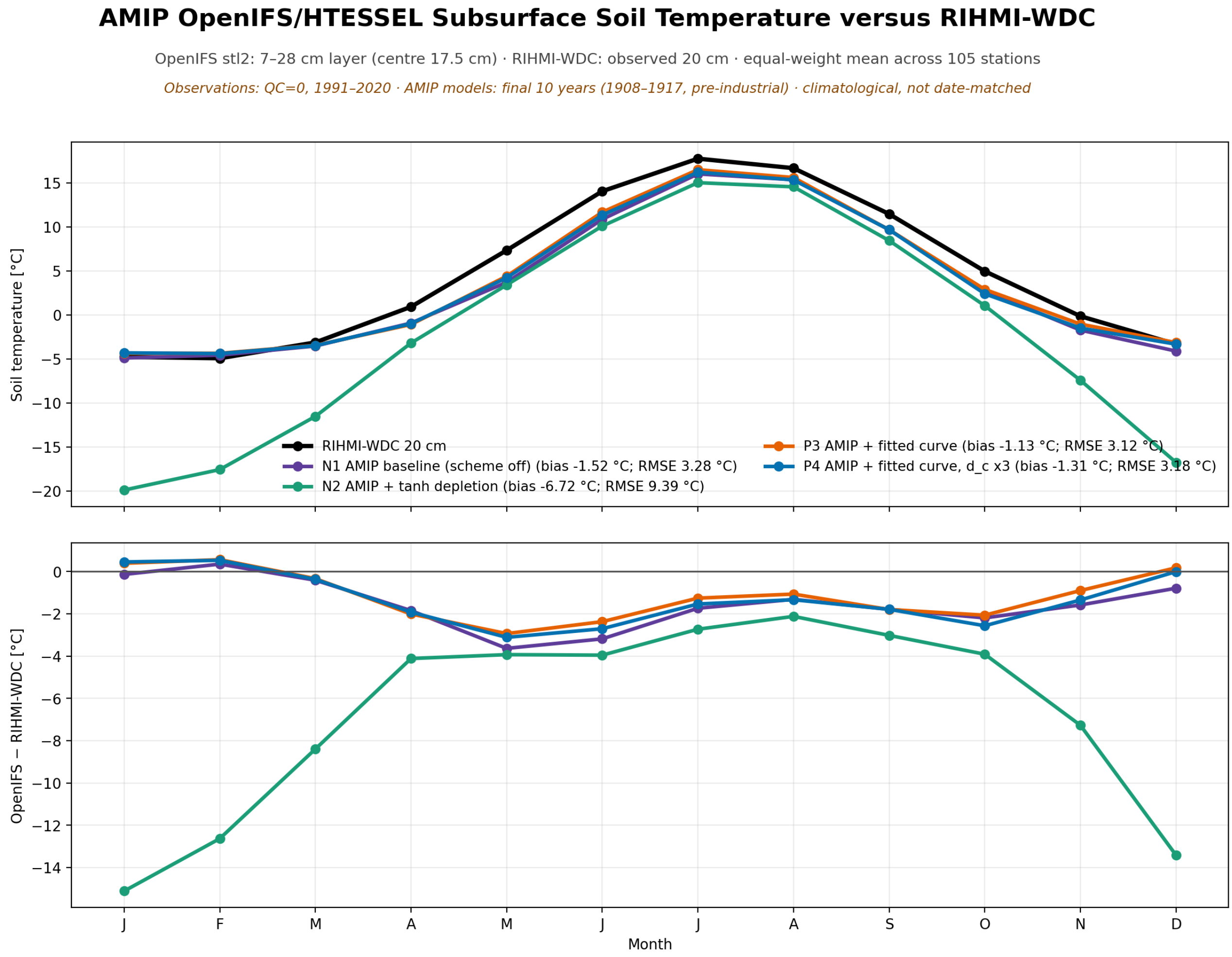

Figure 3: AMIP soil temperature against 105 RIHMI-WDC stations (`stl2` 7–28 cm vs observed 20 cm, QC=0). The falsified tanh run (green) peels away from October and reaches −15 °C in January. **The adopted runs lie on the scheme-off baseline all winter and slightly above it**, and beat it on every metric — bias −1.13 vs −1.52, RMSE 3.12 vs 3.28. **The remaining message of this figure is the April–September gap**: every run including scheme-off tracks the observations through Dec–Mar and then falls 2–3.5 °C short all growing season, identically. No snow scheme can touch that — it is the global tropospheric cold bias of §6.

4.4 Closed by measurement

Three further Siberian routes are shut, which matters because each was once the leading candidate. **Land surface albedo** is not the explanation for the universal bias: removing 74 % of a directly measured excess bought 0.089 K of ∼0.7 K, so a perfect fix buys ∼0.12 K, one sixth of the target. **Skin conductivity** is falsified twice over — raising exposed-snow λ_{sk} made DJF monotonically *worse* (−2.834 at 15, −3.226 at 25), and the moss/organic-layer proxy was worse still (−0.168 and −0.314 K vs G4). **The boreal budget is nearly spent**: Siberia's −2.58 K JJA is ∼0.7 universal (present over prescribed-SST ocean too), ∼0.3 orographic and 1.3–1.6 boreal-specific, and G4 has taken ∼0.95 of the last. The remainder is not reachable by a boreal-indexed lever — which is why F5, combining four boreal levers, equals F4 alone to 0.003 K.

5 Why the Southern Ocean and Siberia can both fit

5.1 The trade-off line is a property of one lever family

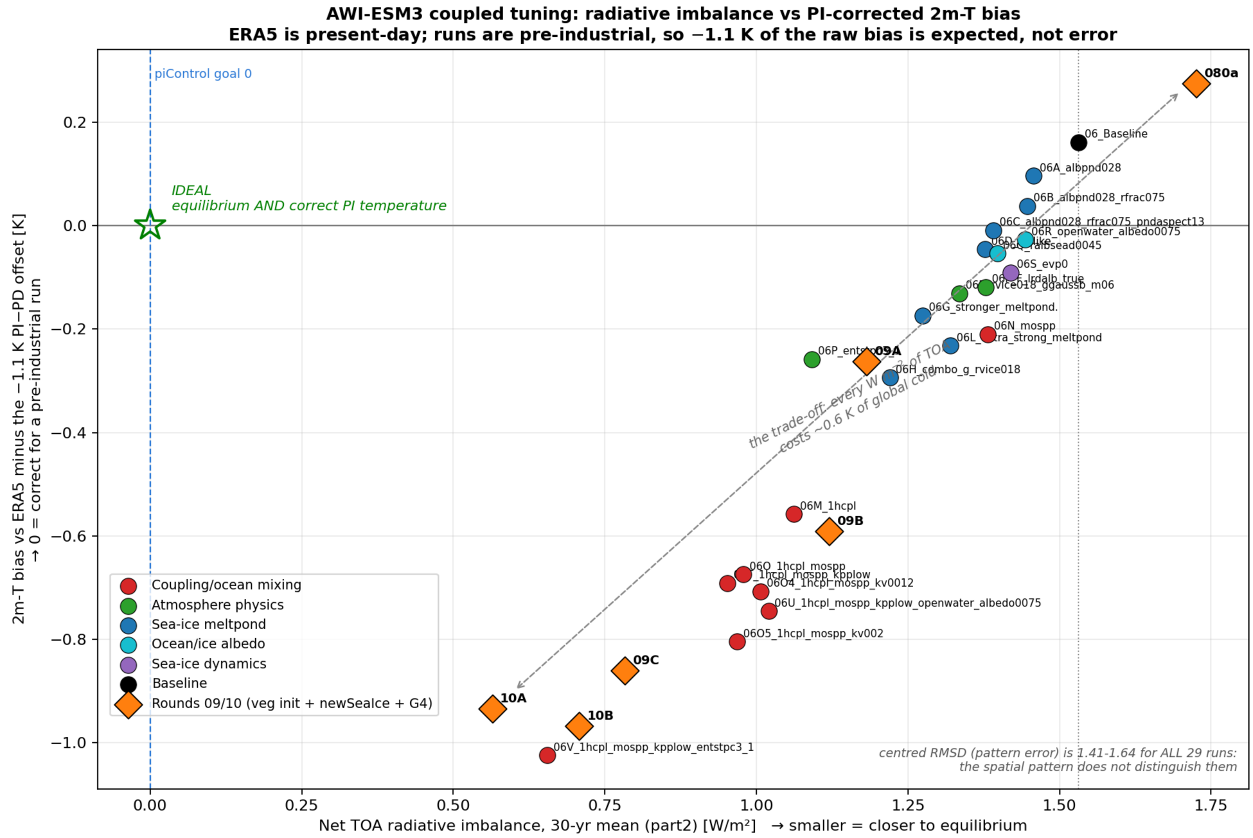

Figure 4: Net TOA imbalance against 2 m-T bias vs ERA5, *corrected for the reference period* (ERA5 is present-day, the runs pre-industrial, ≈1.1 K apart, so 0 is correct for a PI run). **The runs lie on a line, not in a cloud**: from 080a at $(1.73, +0.28)$ to 10A at $(0.57, -0.93)$, every $\mathrm{W\,m^{-2}}$ of TOA improvement has cost ≈0.6 K of global cold. **And the spatial pattern is not what distinguishes them** — centred RMSD, $\sqrt{\mathrm{RMSD}^2 - \mathrm{bias}^2}$, is 1.41–1.64 K across all 29 runs, a ±8 % spread against a raw-RMSD spread of 64 %. The entire apparent skill difference between these runs is one number: the global mean.

Two facts about that figure decide the strategy.

(i) The slope belongs to the ocean-mixing family, not to the model. Every point on the steep part of the line is a coupling/mixing variant. Those levers lower TOA by cooling the planet, which is why they trace a line rather than a cloud, and why CMPI degrades everywhere including regions an ocean-mixing parameter has no business reaching.

(ii) The pattern error is flat. If the runs differed in *where* they are wrong, the centred RMSD would spread; it does not. So the coupled "cold bias" objective is a global-mean problem, and a lever that fixes a regional radiative error without changing the global mean temperature moves the x axis *without* moving the y axis. That is a corner of the plot nothing currently occupies, and it is reachable.

5.2 Disjoint levers superpose — measured, not assumed

G1 = F4 (boreal surface exchange) + D2b (SO ice nucleation) was the first pairing whose components are disjoint in both process and geography. Because two earlier combinations had got superposition wrong *in sign*, it was measured rather than assumed:

metric	predicted (sum)	measured	
Siberia JJA T2m [K]	+0.527	**+0.521**	additive
SO surface-SW RMSE	−2.131	−2.067	additive
SO TOA SW CRE [$\mathrm{W\,m^{-2}}$]	−2.719	−2.840	additive
global net TOA [$\mathrm{W\,m^{-2}}$]	−0.188	−0.195	additive

Additive to within noise on every metric. The plausible reason it holds here and failed for the earlier pairs: F4 acts on a land surface flux in boreal summer, D2b on ice nucleation over ocean below 700 hPa, whereas the failed combinations modified the *same* cloud scheme in the *same* regime.

5.3 The recipe

1. **Buy the imbalance with Southern-Ocean cloud, not with global cooling.** The SO deficit is a reflection error over ocean; correcting it adds reflected SW where the model is warm-biased. It lowers TOA *and* cools a region that should be cooler, so it moves both axes the right way instead of trading them.
2. **Buy Siberia with boreal-indexed surface exchange and the snow scheme.** Both are land-only, summer-weighted, and by construction cannot touch the tropics or the ocean.
3. **Stop escalating ocean mixing.** There is only 0.18 W m^{-2} of coupled-state term left in 10A anyway, and buying it costs ≈0.1 K of global cold for a term the atmosphere is now known to dominate 4:1.
4. **Score every lever regionally.** The per-band TOA decomposition existed only for the control until recently — the energy target was tracked by one global number with no idea which band any lever was moving. It is now in the standard table for every run, and it is what makes the disjointness in (1)–(2) verifiable rather than asserted.

The G-series is the existence proof: G1/G4 improves the Southern Ocean SW RMSE by 30 %, warms Siberian JJA by ≈1 K, leaves the tropics inside +0.12, and lowers global net TOA. That combination is off the trade-off line.

6 The shared limiter: the global tropospheric cold bias

Model minus ERA5 on the model's own pressure levels, `amip_presentday` vs ERA5 1990–2014 so the comparison is period-clean:

hPa	Siberia	Global	**Siberia-specific**
1000	−1.90	−0.73	**−1.17**
925	−2.17	−0.65	**−1.52**
850	−1.99	−0.99	**−1.00**
700	−1.22	−1.15	−0.07
500	−1.67	−1.49	−0.18
300	−2.32	−2.22	−0.10

This separates two problems that had been conflated for most of the campaign. The boreal-specific excess lives entirely in the bottom three levels — the signature a surface-exchange problem should have, which is what validates the F/G framing. **Above 700 hPa Siberia merely shares a global tropospheric cold bias of 0.7 to 2.2 K that no boreal lever will ever touch**, and which has never been worked on in 50 runs.

Three separate observations now converge on it, which is why it belongs in a summary rather than an appendix:

- It is the residual in the Siberian budget — the ∼0.7 K "universal" component present over prescribed-SST ocean and under every one of the 20 surface types.
- It is what the RIHMI station network sees as the 2–3.5 °C April–September soil deficit, identical in every AMIP run including scheme-off.
- It is the natural suspect for the residual radiative error. A cold troposphere under-emits; the model's OLR is −0.74 W m^{-2} below CERES while its absorbed SW is only −0.37 below, so relative to observation the model's positive imbalance *is* its OLR deficit.

That last point needs a caveat and it is worth stating precisely, because it is the next thing to test. A Planck scaling of a ≈1.5 K cold emitting layer predicts an OLR deficit of order 5 W m^{-2}, not 0.74. Either the cold bias sits where it does not radiate to space, or several W m^{-2} of compensating error are present. The comparison as it stands cannot distinguish these, because **the TOA decomposition was measured on a PI run against present-day CERES while the tropospheric bias was measured on the present-day run against ERA5**. The clean version costs no model time: `amip_presentday` already exists and already reads net TOA +2.20 against CERES's +0.97. Running the band decomposition on *that* arm re-bases every energy target in the campaign onto a period-clean footing and tests the OLR link directly.

7 Closed by measurement

Only the mechanisms whose death still changes a decision. The full list of 29, with the measurement that killed each, is § "Falsified claims, kept" in `report.tex`.

claim	what killed it
Aerosol is ∼13 % of the SO clear-sky excess	Inferred from AOD vs MISR. Measured by removing all anthropogenic aerosol: $-0.116\,\mathrm{W\,m^{-2}}$, $t = -18.9$ — six times smaller. The unexplained SO residual grew to ∼1.9.
The band contributions show no region carrying an anomalous imbalance	Artefact of comparing bands to each other, not to CERES. Differenced properly, the Southern Ocean carries most of it.
The tanh snow depletion is a valid subgrid formulation	Its codomain is open at 1; snow courses report exactly 10/10 in 99.74 % of DJF surveys. -20.2 K against 174 676 station observations where the scheme *off* is right to +0.8.
The June albedo bias is worth $\sim 5.6\,\mathrm{W\,m^{-2}}$ of boreal warming	Fixing it to satellite accuracy gained +0.094 K — four times less efficient per watt than the vegetation route.
Skin conductivity is the Siberian winter route	DJF got monotonically *worse*: -2.834 at λ_{sk}=15, -3.226 at 25. The moss proxy failed the same way.
`RVRSMIN` saturates near 1000	500/1000/2000 → $+0.336/+0.521/+0.876$: increments *accelerate*. 1000 is a damage cap, not a knee.
About half the boreal bias is the reference period (+1.12 K)	Indirect HadCRUT5 chain; measured directly on the model's own present-day arm it is +0.42 K, wrong by 2.7×.
Siberian JJA net SW is $33\,\mathrm{W\,m^{-2}}$ too low	Measured against CRUNCEP3, itself $\approx 21\,\mathrm{W\,m^{-2}}$ too bright. Against CERES the deficit is 11, split 7.0 cloud + 5.5 albedo.
The scale dependence of the snow floor is a measurable power law	The ladder confirms the sign but the values plateau beyond 133 km and the per-cell fit disagrees 3× with the seasonal fit. Kept as a two-anchor interpolation, labelled as such.
The HR piControl goal of -0.16 is the target	Inherited, not derived. The budget closes to $0.02\,\mathrm{W\,m^{-2}}$, so the goal is 0.

Two methodological lessons are worth as much as the physics. A lever was once promoted on a 4-year window at +0.502 K and re-measured at −0.038 over 44 years — pure noise, after two runs had already been built on it; the run×year ANOVA thresholds exist because of that. And JJA alone was evaluated for eleven rounds while the model's original complaint was a *cold-season* bias, which hid a −0.72 K DJF penalty in the then-best lever. Both failure modes are now structurally prevented by a single `evaluate.sh` that runs the whole protocol, because the way to make a guardrail effective is to make skipping it require effort.

8 The path forward

Ranked by expected value per unit of model time. Each carries its falsifier, stated in advance.

1. The Southern Ocean cloud-area deficit (∼6 pp). The largest unexploited term in the campaign: $+5.81\,\mathrm{W\,m^{-2}}$ in-band, +0.57 global on its own, untouched by all 34 evaluated levers, and now with a *larger* unexplained residual than before because aerosol was measured out of it. The live mechanism is supercooled liquid: RH is +5.8 to +6.6 % too high through the mixed-phase layer while reflection is too low, and D2b — the only lever that has moved the pattern — works by ice nucleation. Continue down that branch, and **score on cloud area, not on CRE**, since amount and opacity are roughly half the error each and a CRE metric cannot tell them apart. *Predicted:* closing the deficit buys $\approx 0.5\,\mathrm{W\,m^{-2}}$ of the imbalance at near-zero global temperature cost. *Falsifier:* if the response also moves the tropics or NH T2m beyond noise, it is a global cloud knob in disguise and must be rejected as A1a was.

2. Re-base the energy target period-cleanly. No model time at all: run the per-band TOA decomposition on the existing `amip_presentday` arm against CERES 2005–2015. That arm reads +2.20 against CERES's +0.97, so the true radiative fault is ≈ +1.23 — twice what the PI framing reports — and every band target in this report currently carries an unquantified epoch offset. It also supplies the direct test of the OLR/tropospheric-cold link of §6. **Do this before designing item 1's runs**, because it may move the target.

3. Diagnose the global tropospheric cold bias. Now the dominant remaining term for land 2 m temperature, the residual in the Siberian budget, the whole of the RIHMI growing-season deficit, and the natural suspect for the residual imbalance. Fifty runs have not touched it, and no lever built so far can. This needs *diagnosis first* — where the column loses its heat, and whether the compensating error implied by the OLR arithmetic is real — not another parameter sweep.

4. Verify K1 + P5 + G4 coupled. Three things AMIP structurally cannot price. Whether 48 % closure of the boreal bias suffices for forest survival, since the coupled model closes a runaway that AMIP deliberately severs. Whether G4's delay of melt-out from 24 to 28 May *harms* forest establishment through later growing-season onset — a real risk sitting exactly where AMIP is blind. And the Arctic energy bill: prescribed SST and ice absorb $1.7\,\mathrm{W\,m^{-2}}$ without responding, and the coupled spin-up will not.

5. The snow-free land albedo residual ($+0.0080$; crops $+0.0204$, semidesert $+0.0169$, bare soil $+0.0145$). The one remaining lever aimed at the ∼0.7 K *universal* bias rather than the boreal one, untouched by all 50 runs. Worth ∼0.12 K — modest, but cheap, and it is namelist-only. *Caveat carried forward:* it must not simply be stacked on the snow levers, because the tundra entry competes for the same Arctic energy budget.

Explicitly not worth further runs: `SWEMIN` optimisation (P3 and P5 differ by 0.19 K in JJA, at or below noise, and the total available May cover error is worth $\approx +0.2$ K); boreal-indexed surface levers beyond G4 (F5 = F4, and the boreal-specific budget is ∼95 % spent); land albedo as an explanation for the universal bias (one sixth of the target, measured).

9 Carried forward

K1 + P5, namelist-only on top of G4, mergeable with EC-Earth as one added line in an ECMWF namelist file — no source divergence to maintain.

component	setting	why
G4	`RVRSMIN`$(3,4) = 1000$, tundra $80 \to 225$	Route A: cloud term $-7.04 \to +1.20$
K1	snow-free land albedo correction	+0.089 K, best global T2m RMSE bar K2
P5	`ECE_SNOW_SCF`$= 3$, `SWEMIN`$= 15$	fitted depletion, every season clean
D2b	`INPSEA`$= 0.2$ gated below 700 hPa	SO SW RMSE -28 %, zero tropical cost

Resolution. The snow scheme carries a preset, $\text{SWEMIN}_{\text{eff}} = \text{clamp}(\text{SWEMIN} \cdot (\Delta x/100)^{0.624}, 3, 120)$, off by default so every run in this report reproduces bit-for-bit. Presets: TL21 94, TL63 48, TL255 26, TCO95 30, TCO399 13, TCO1279 6, TCO4000 3; below 2 km the scheme disables itself with a log line, because at that scale the model resolves the patches a subgrid curve would double-count. **It is an interpolation between two measured anchors, not a law** — a new resolution needs validation, not just a `DXKM` value.

Reproduction. `scripts/analysis/evaluate.sh` runs the full protocol — seasonal ANOVA against the 44-yr thresholds, the Siberian box, the per-band TOA decomposition, the global guardrails — for any run in `scripts/analysis/runs.py`. Every figure and table in this summary traces to a script in `scripts/`, and the run-by-run parameter record is `notes/RUNS_AND_PARAMETERS.md`.